

SKYLAR GUTMAN

Embedded Systems · Hardware-Software Co-Design · Robotics

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EDUCATION

B.S. Computer Engineering · Florida Atlantic University, Boca Raton, FL · May 2026 · GPA 3.4

Relevant Coursework: Microprocessors · Computer Architecture · Advanced FPGA Design · Autonomous Robots · Computer Operating Systems · Circuits & Electronics · iOS Mobile Apps · Software Engineering

PROJECTS & LAB EXPERIENCE

ROS2 RedBot Mobile Robot Platform | *Personal / FAU Robotics* 2024 – 2025

- Designed and implemented a full ROS2 Humble package (redbot_bridge) on Raspberry Pi 4, enabling serial communication between a SparkFun RedBot Arduino and ROS2 nodes.
- Built differential drive control via Twist messages, IMU/encoder publisher nodes, and end-to-end teleoperation; validated with rviz2 visualization.

Cart-Pendulum Control System | *FAU — Autonomous Robots / Controls* 2025 – 2026

- Developed MATLAB/Simulink models for encoder-based state estimation on a cart-pendulum system; implemented embedded code generation targeting real-time hardware.
- Tuned PID and LQR controllers; debugged signal conditioning and model-to-hardware deployment pipeline.

Conductive-Fingertip Gesture Glove | *Personal Project* 2024

- Built a BLE wearable on ESP32 that detected finger-combination gestures via conductive fingertips and triggered modular Python actions: Twilio calls, email, and media playback.
- Designed the hardware sensing circuit, BLE GATT profile, and host-side Python dispatcher from scratch.

CNC Router — Full Sheet Plywood Capacity | *Personal Project* 2023

- Designed and fabricated a full-sheet CNC router with water-cooled spindle; engineered the chain-drive gantry system and tuned motion controller firmware.
- Used SolidWorks for mechanical design, then built, wired, and commissioned the machine end-to-end.

Jewelry Tumbler — Benchtop Prototype | *Personal Project* 2023

- Designed a motorized jewelry polishing tumbler in SolidWorks; produced all structural components on a 3D printer and assembled the drivetrain.
- Documented the build and published to YouTube: youtube.com/shorts/doUxGtvKBY8

IoT Security — AES-256 Encrypted MQTT on AWS IoT Core | *Personal Project* 2024

- Built an encrypted MQTT publisher in Python (paho-mqtt + pycryptodome) targeting AWS IoT Core; explored RTMP/HLS/WebRTC for live Raspberry Pi video streaming.

Digital Systems Lab — Artix 7 FPGA | *Florida Atlantic University* | Boca Raton, FL 2024 – 2026

- Implemented combinational and sequential logic designs in VHDL on the Artix 7 FPGA; verified timing and functional correctness with ModelSim and oscilloscope probing.

Engineering Lab — Circuits & Electronics | *Florida Atlantic University* | Boca Raton, FL 2024 – 2025

- Built and characterized RC, op-amp, and MOSFET circuits on prototype boards using Tektronix oscilloscopes, function generators, and DMMs for signal analysis.

RepRapPro Mendel 3D Printer Build | *Personal Project* 2022

- Assembled a full FDM printer from printed parts and threaded rod; sourced only components that could not be fabricated in-house. First-hand experience with motion systems, extruder calibration, and firmware tuning.

PROFESSIONAL EXPERIENCE

Project Manager & IT Lead | *Ultimate Environments (General Contractor)* | Boca Raton, FL Mar 2020 – May 2025

- Read architectural and MEP drawings to plan construction schedules and coordinate subcontractors for commercial retail fit-outs.
- Managed end-to-end project delivery — estimating, scheduling, procurement, and punch-list — for retail stores ready to open on day one.
- Administered IT infrastructure for all employees: email, network security, server and workstation maintenance.

Software Engineer | *StarStar Mobile* | Pompano Beach, FL 2015 – Mar 2020

- Designed and implemented a web-based CMS for mobile content delivery using PHP and the Symfony framework; system was moderated by business users at scale.

Owner / IT Consultant | *Signature Computer Service* | Coral Springs, FL 1998 – Present

- Provide network, hardware, and software services to local businesses; manage client relationships, scoping, and delivery as a sole operator.

CERTIFICATIONS

FAA Part 107 Remote Pilot Certificate · *Federal Aviation Administration*

- Certified for commercial unmanned aircraft system (UAS/drone) operations in controlled and uncontrolled airspace; knowledge of airspace classification, weather, RF systems, and FAA safety regulations.

TECHNICAL SKILLS

Hardware: ESP32 · Raspberry Pi 4 · Arduino Uno · TI MSP430 · Artix 7 FPGA · Tektronix Oscilloscope · DMM · Function Generator

Embedded / Firmware: C++ · VHDL · Serial/I²C/SPI · BLE GATT · PWM Motor Control · FreeRTOS concepts

Robotics & Simulation: ROS2 Humble · MATLAB/Simulink · NVIDIA Isaac Sim · rviz2 · Differential Drive

Languages: C++ · Python · VHDL · PHP · MySQL · JavaScript · HTML

Software & Tools: SolidWorks · AutoCAD · PSpice · Git · JetBrains IDEs · Linux · VS Code

Protocols & Comms: I²C · SPI · UART · BLE · MQTT · ROS2 Topics/Services

Fabrication: 3D Printing (FDM) · CNC Machining · SolidWorks Mechanical Design

UAS / Drone: FAA Part 107 Certified · Airspace Classification · RF Systems · UAS Safety Regulations